# Mekala Dheekshitha

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 7.16 |
| Sri Chaitanya Junior College | Class XII (Telangana State Board of Intermediate Education) | 2023 | 98.00% |
| Sri Chaitanya High School | Class X (Telangana State Board of Secondary Education) | 2021 | 100.00% |

## POSITIONS OF RESPONSIBILITY

**OCS Placement Coordinator**

Pitched companies for Placement opportunities and managed end-to-end communication between recruiters and students.

Coordinated Online Assessments (OAs), ensuring seamless scheduling, candidate communication, and execution.

**OCS Internship Coordinator**

Pitched companies for internship opportunities and managed end-to-end communication between recruiters and students.

Coordinated Online Assessments (OAs), ensuring seamless scheduling, candidate communication, and execution.

**Class Representative (CR)|B.Tech 2023 batch:**

Acted as liaison between faculty and students to communicate academic updates and resolve issues.

**Events Head, Milan**

Led a 10+ member team in organizing cultural events for Milan, the inter-hostel fest at IITH.

Managed end-to-end planning, budgeting, and execution of major flagship events.

**E-Cell Corporate Relations and Finance Manager:**

Handled corporate communications and built partnerships with startups and companies for E-Cell events and initiatives.

Coordinated sponsorship proposals and follow-ups for securing event funding.

## PROJECTS

**Graph | Multi-Objective Route Optimization Engine**

- Built a real-time multi-modal travel planner using **public TSRTC bus-stop/timetable and real-time metro station data**, optimizing the Fastest, Cheapest, and Best-Value routes.
- Built a modular agent pipeline integrating real-time Maps, Weather, Safety, Ride-Hailing, and Public Transport APIs with ETA and dynamic cost estimation using **Dijkstra algorithm.**
- Engineered a weighted route-ranking framework using predicted waiting buffers, transfer penalties, walking constraints, and dynamic edge weightsforrealistic travelrecommendations.

**Reservoir Pre-release Operations Using Weather Forecasting**

- Conducted research under the guidance of Dr. R. Maheswaran on reservoir pre-release using weather forecast data.

- Analyzed rainfall and inflow forecasts for single and multi-reservoir system management.
- Evaluated **flood mitigation and water storage strategies** under varying forecast scenarios.
- Studied the effects of forecast uncertainty on reservoir operation decisions.

**Transportation Engineering Research Project**

- Conducted research on **driver speed behaviour at merging sections** under the guidance of Prof. Digvijay.
- Applied dynamic clustering and traffic flow analysis techniques to identify driver behaviour patterns.
- Utilized Python for data processing, analysis, and visualization of traffic datasets.
- Developed analytical graphs and insights to evaluate speed variations and merging characteristics.

**Marketing Research & Questionnaire Design**

- Spearheaded the **design of a consumer preference questionnaire** for [Weekend Holiday trip service at IITH], coordinating team efforts to ensure logical flow and minimize response bias.
- Delivered a strategic defense of the survey methodology to faculty, effectively managing a critical 10-minute Q&A session on research design

## SKILLS

**Programming Languages** : C, C++, Python, HTML, CSS, MS Excel

**Libraries** : NumPy, Pandas, Matplotlib, Seaborn, FastAPI

**Tools** : Visual Studio Code, Google Colab, Jupyter Notebook, Fusion 360, Tableau, GitHub

**Certificates and courses: C**omplete Data science and ML Bootcamp, Complete DSA Bootcamp

## RELEVANT COURSES

**Mathematics** :Statistics,Probability,Calculus,Linear Algebra,Differential Equations,Complex variables,Transform Techniques

**Programming courses** : Intro to Programming(C), Python, SQL

**core courses**: Geotechnical Engineering, Transportation Engineering, Hydraulics Engineering, Environmental Engineering

**Electrical Engineering** : Basic Electrical engineering, Electrical circuits lab

**Mechanics** : Engineering Mechanics, Fluid Mechanics, Solid Mechanics

**Others** : Engineering Drawing, Digital Fabrication, Modern Physics, Environmental Chemistry

## EXTRACURRICULAR

Completed beginner level Japanese language course to explore global cultures and professional opportunities

Won second Prize in Waste Management Hackathon in Elan & Vision (IITH)

Won Gold medal in Billiards (Milan 2024, 2025)

Interests : Classical Dance, Crochet, Gym, Billiards